MISSION27

NASA Scientific Data Sources & AI-Assisted Development

Project: MISSION27 – Train on Earth. Survive the Moon

Team: CREW-M27

Challenge: NASA Space Apps Challenge 2026 – Build a Junior Astronaut Mission Trainer

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MISSION27 incorporates NASA scientific datasets, archived observations, educational resources, and 3D assets to create an interactive lunar exploration experience. The following table summarizes the 12 NASA resources, their purposes, and how they are used in the game.

Data table:

No.	NASA Data Source	Purpose	Integration in MISSION27	Official Source
1	NASA OSDR-766 (Space-Grown Tomato Experiment)	Space agriculture, plant growth, and LED lighting analysis	NASA's 2026 tomato experiment data supports greenhouse training, allowing players to compare red-rich (90:10) and blue-rich (50:50) lighting treatments and simulate crop-management decisions.	NASA OSDR-766
2	NASA RadLab (Radiation Dosimetry Database)	Space radiation monitoring and exposure analysis	Historical ISS radiation measurements are used in the Radiation Safety Lab, where players analyze dose-rate graphs and practice radiation-safety decision-making.	NASA RadLab

3	NASA JPL Horizons (Solar Position Data)	Solar position calculations and energy optimization	NASA-calculated solar elevation and azimuth data support Earth Training's Solar Energy Lab, allowing players to adjust simulated solar-panel orientation and optimize energy generation.	NASA JPL Horizons
4	LRO – LOLA (Lunar Orbiter Laser Altimeter)	Lunar terrain modeling and elevation analysis	Processed elevation data informs Shackleton-region terrain, rover navigation, and slope assessment.	NASA LRO
5	NASA DONKI (Space Weather Database)	Solar particle events and radiation safety	Archived May 11, 2024 solar event informs radiation emergencies, crew protection, and sheltering decisions.	NASA DONKI
6	NASA CRaTER (Cosmic Ray Telescope)	Radiation measurement and analysis	Archived July 2, 2009 detector measurements support radiation inspection and scientific interpretation.	NASA PDS
7	NASA Lunar South Pole Environment	Lunar illumination and solar energy	Scientific references inform sunlight availability, solar-power generation, and energy-management challenges.	NASA Moonbase Environment
8	NASA ECLSS (Environmental Control and Life Support System)	Water recycling and sustainability	NASA's 98% water recovery milestone informs water-management and life-support activities.	NASA ECLSS

9	NASA OSDR – OSD-269	Space agriculture and plant growth	Research on plant lighting informs greenhouse operations, crop stress, and red-blue-green lighting decisions.	NASA OSDR
10	NASA LRO – LEND	Lunar hydrogen exploration	Published neutron-suppression research informs lunar survey activities and hydrogen exploration decisions.	NASA LEND Research
11	NASA EMU 3D Model (Extravehicular Mobility Unit)	Astronaut equipment and EVA preparation	NASA astronaut suit model supports equipment inspection, oxygen checks, cooling, and battery preparation.	NASA 3D Resources
12	NASA EVA Database	Spacewalk duration and battery planning	Apollo 17's first EVA duration (7 hours, 12 minutes) provides a historical benchmark for battery-planning activities.	NASA EVA Database
13	NASA JPL – LSOS (Lunar Surface Operations Simulator)	Rover mobility and lunar operations	NASA research provides design inspiration for rover movement, terrain-based driving, and safe exploration decisions.	NASA JPL
14	NASA DSCOVR – EPIC (Earth Polychromatic Imaging Camera)	Earth observation and photography	Archived May 10, 2024 Earth imagery supports the Earth-photography activity with camera orientation, focus, and exposure.	NASA EPIC API

15	NASA Apollo Lunar Sample References	Lunar geology and sample classification	Historical Apollo sample photographs support lunar material comparison, sample collection, and science-lab activities.	NASA Sample Documentation / LPI Sample Atlas

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2. AI Tools Used in DevelopmentMISSION27 was developed through a combination of human-led project planning, scientific research, programming, 3D game development, and AI-assisted software development.

The following tools contributed to the project's development process.

2.1 ChatGPT – OpenAI

Website: https://chatgpt.com/

Role: Planning, scientific-content organization, and debugging.

ChatGPT was used to assist with developing and refining MISSION27, including:

- Brainstorming game mechanics and interactive learning activities.
- Structuring astronaut training and lunar mission scenarios.
- Organizing NASA scientific references into educational gameplay.
- Assisting with React, TypeScript, and 3D development concepts.
- Assisting with technical explanations and development documentation.

ChatGPT functioned as an AI-assisted development and problem-solving tool rather than an autonomous project creator.

2.2 Lovable – AI-Powered Application Development

Website: https://lovable.dev/

Role: Main browser-based game development environment.

Lovable was used to implement and iteratively improve the interactive MISSION27 web application.

Its contributions included:

- Generating and modifying frontend application code.
- Building interactive interfaces and game components.

- Implementing the lunar exploration environment.
- Integrating astronaut movement, mission objectives, and resource-management interactions.
- Refining the rover model and driving controls.
- Improving mission alerts, scoring interfaces, and mobile responsiveness.
- Supporting testing, debugging, and deployment workflows.

The team provided project requirements, reviewed generated implementations, reported problems, and directed revisions throughout development.

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3. Development Transparency

MISSION27 is a human-directed educational game developed with AI-assisted programming tools and NASA scientific resources.

The team was responsible for defining the project concept, identifying educational objectives, selecting scientific references, designing learning activities, evaluating game mechanics, and requesting improvements throughout development.

AI tools supported implementation, programming assistance, debugging, and documentation. NASA resources provided scientific references and selected datasets, while the mission scenarios, scoring systems, and interactive learning mechanisms were implemented as educational game simulations.

Scientific attribution: MISSION27 is an independent educational project developed for the NASA Space Apps Challenge. It is not an official NASA product and is not affiliated with or endorsed by NASA.